

Chen-Yu Hu

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Research Profile

M.S. Computer Science researcher studying multilingual and multimodal language understanding, spoken dialogue systems, and visual grounding. Research experience includes task and benchmark formulation, annotation design, multimodal and LLM-based evaluation pipelines, and error analysis across speech, text, and vision. First author of a manuscript currently under review in spoken dialogue state tracking.

Research Interests

Multimodal language understanding; spoken dialogue systems; dialogue state tracking; ASR-LLM integration; visual grounding and reference resolution; evaluation and benchmark design; human-robot interaction.

Education

Institute of Science Tokyo , Tokyo, Japan M.S. in Computer Science, Yoshino Lab	Apr 2025–Mar 2027 (expected)
- Research topic: grounding spoken expressions to handwritten mathematical content. - Grade: 91.2/100.	
Tokyo Institute of Technology , Tokyo, Japan Exchange Student in Computer Science; grade: 92/100	Oct 2022–Aug 2023
National Tsing Hua University , Hsinchu, Taiwan B.S. in Computer Science and Engineering; GPA: 4.01/4.30	Sep 2019–Aug 2023

Research Experience

Incoming Research Assistant , RIKEN, Japan	Nov 2026–Mar 2027
Research Assistant , Yoshino Lab Institute of Science Tokyo, Tokyo, Japan	Aug 2025–Apr 2026
- Conducted first-author research on grounding referring expressions in English and Japanese instructional mathematics videos to handwritten equations, subexpression spans, terms, figures, and figure parts. - Developed a controlled mention-to-box ranking task and contributed to annotation design, data curation, benchmark construction, and automated consistency validation. - Established exact-ID and ranking evaluation protocols and analyzed ambiguity arising from discourse context, overlapping visual candidates, and target granularity. - Evaluated non-learned, frozen dual-encoder, and vision-language model baselines, including the effect of human-reviewed candidate-level LaTeX metadata.	
Student Research Intern , NexaScience Inc., Tokyo, Japan	Sep 2025–Dec 2025
- Built a reproducible spoken dialogue understanding pipeline integrating ASR, LLM-based dialogue state tracking, and automatic evaluation. - Designed experiments comparing ontology-guided ASR post-editing with direct expansion of database-backed slot candidates in a modular ASR-DST architecture. - Analyzed joint goal accuracy, slot error rate, canonical-name recovery, and extra slot predictions under noisy ASR input. - Led the resulting first-author ICASSP 2027 submission, contributing experiment design, pipeline implementation, error analysis, and manuscript preparation.	

Manuscript Under Review

- Chen-Yu Hu**, Yuzheng Xu, Tadashi Kozuno, Yoshitaka Ushiku, and Koichiro Yoshino. “Comparing Post-Editing and Slot Candidate Expansion for Spoken Dialogue State Tracking.” Submitted to IEEE ICASSP 2027; under review.

Selected Research Presentations

- **Chen-Yu Hu**, Takuto Asakura, and Koichiro Yoshino. “Linking Pronouns in Dialogue with Whiteboard Symbols: Towards Confusion Detection in Math Collaboration.” 9th International Workshop on Symbolic-Neural Learning, Oct 2025.
- **Chen-Yu Hu**, Takuto Asakura, and Koichiro Yoshino. Same title. 16th Dialogue System Symposium, Nov 2025.
- **Chen-Yu Hu**, Takuto Asakura, and Koichiro Yoshino. “Towards Dialogue Structure Analysis of Math Collaboration Using Whiteboards.” 32nd Annual Meeting of the Association for Natural Language Processing, Mar 2026.

Academic Project

- Facial Features Transferring**, Senior Project
National Tsing Hua University, Taiwan
- Feb 2022–Sep 2022
- Built a facial-feature transfer and image-synthesis pipeline using StyleGAN2 and pixel2Style2pixel.
 - Designed the system architecture and implemented geometric warping and color correction using Python and OpenCV.
 - Integrated MediaPipe 3D face-mesh extraction to improve alignment and visual realism.
 - Selected as a finalist in the NTHU Senior Project Competition.

Industry Experience

- Software Engineer Intern**, Dykoo Inc., Taichung, Taiwan
- Jan 2024–Apr 2024
- Built features for an eCommerce order-management application using SvelteKit and TypeScript.
 - Designed order-splitting algorithms and data-processing workflows using Python and Pandas for warehouse picking and inventory tracking.
 - Automated a half-day manual order-organization workflow into a brief run-and-review process for a company user.
 - Built Excel parsing and Firebase/Firestore integration workflows using Cloud Functions.

Awards and Scholarships

- Scholarship Recipient**, Watanuki International Scholarship Foundation
- Apr 2026–Apr 2027
- Finalist**, NTHU Senior Project Competition
- Dec 2022
- Academic Excellence Award** (Top 5%), NTHU
- Spring 2022
- JASSO Scholarship**, Japan
- Oct 2022–Aug 2023
- Exchange Student Scholarship** for academic performance
- Oct 2022–Aug 2023

Skills and Languages

- Research:** Multimodal learning, visual grounding, NLP, ASR, dialogue state tracking, LLM evaluation, experimental analysis, annotation and benchmark design.
- Programming:** Python, C/C++ fundamentals, TypeScript, SQL.
- Frameworks and Tools:** PyTorch, TensorFlow, OpenCV, MediaPipe, Firebase/Firestore, Git, LaTeX.
- Languages:** Mandarin and Taiwanese (native), English (IELTS 7.5; TOEIC 910), Japanese (JLPT N1).

Selected Coursework

- Data Structures, Algorithms, Linear Algebra, Operating Systems, Machine Learning, Computer Vision, Natural Language Processing, and Image Processing.